

Time-Series Forecasting with Recurrent Neural Networks in PyTorch

Benchmarking vanilla RNN, LSTM, and GRU architectures on a decade of daily temperature data using GPU-accelerated PyTorch.

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Overview

This project implements and compares three recurrent neural network architectures for one-step-ahead time-series forecasting. The dataset is the Daily Minimum Temperatures series recorded in Melbourne from 1981 to 1990, comprising 3,650 daily observations. A vanilla RNN, an LSTM, and a GRU are each trained with GPU acceleration in PyTorch and evaluated on a held-out test set using RMSE and MAE.

Goals

Determine how well recurrent architectures capture the seasonality of a real-world temperature series and assess whether gated units (LSTM, GRU) offer a meaningful advantage over a plain RNN on a short-range prediction task.

Objectives

- Split the series chronologically (80/20) and apply MinMax scaling fitted on the training portion only to prevent data leakage.
- Construct 30-day sliding windows as model inputs with the next day's temperature as the prediction target.
- Implement a configurable recurrent model class supporting RNN, LSTM, and GRU cells with a linear output head.
- Train all three variants on the GPU and record test RMSE and MAE for a fair side-by-side comparison.
- Visualise training-loss curves, an RMSE comparison bar chart, and the best model's forecast overlaid on the actual series.

Approach

Data is preprocessed with a chronological split and MinMax scaling to keep the evaluation realistic. Sliding windows of 30 days feed into a shared model architecture where only the cell type changes, ensuring a controlled comparison. Each model is trained with the Adam optimiser on a GPU (RTX 3080) and the final test metrics are computed after training completes.

Outcomes

- All three models achieved test RMSE values near 2.23-2.29 degrees C, demonstrating reliable learning of the seasonal pattern.
- Vanilla RNN reached RMSE 2.234 degrees C and MAE 1.759 degrees C; LSTM matched it at

RMSE 2.234 degrees C and MAE 1.762 degrees C.

- GRU scored RMSE 2.287 degrees C and MAE 1.800 degrees C, marginally behind the simpler architectures.
- The near-identical scores confirm that gating provides little benefit for one-step-ahead prediction of a smooth, strongly seasonal signal where only recent context matters.